

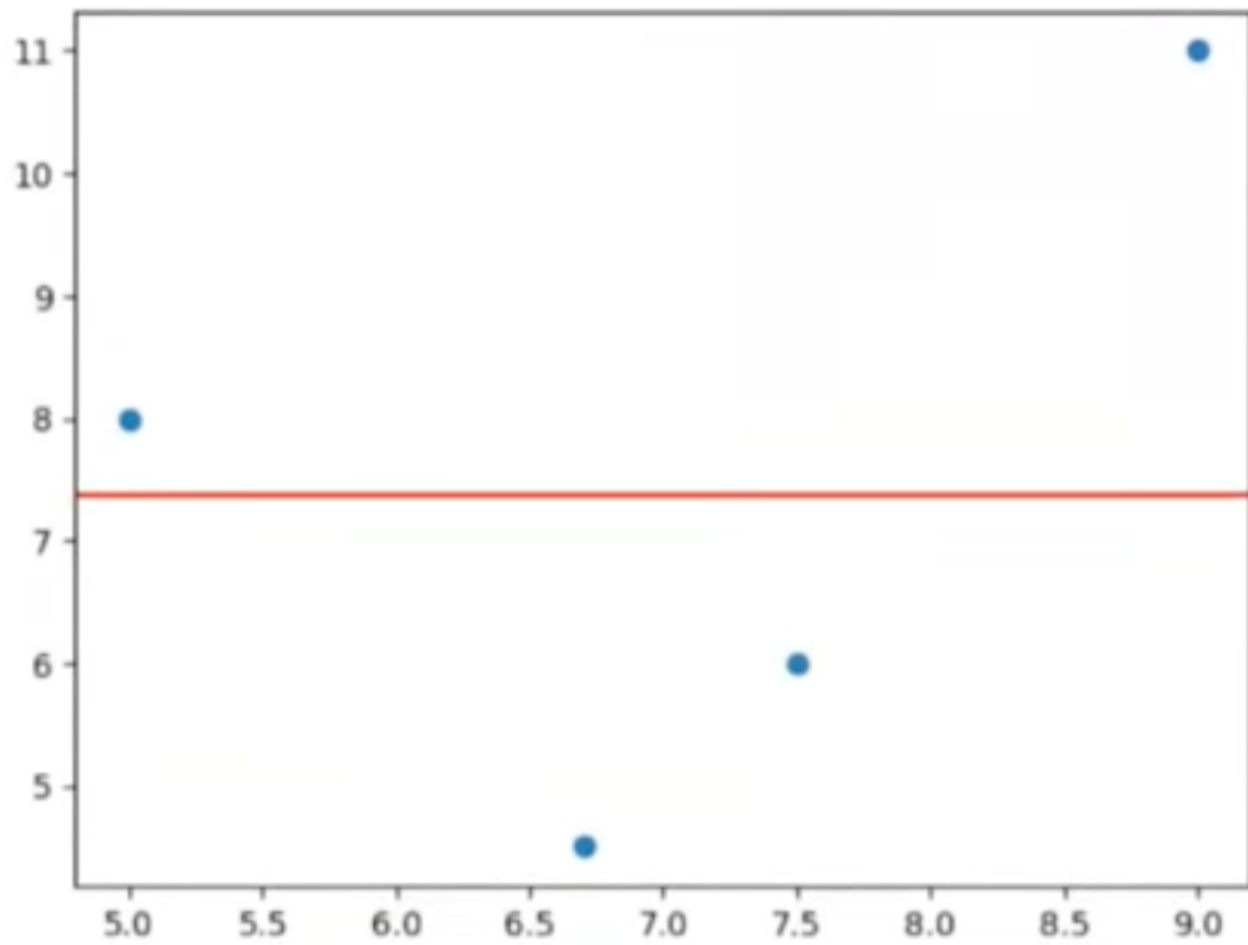
XGBoost For Regression

cgpa	package
6.7	4.5
9.0	11.0
7.5	6.0
5.0	8.0

Stage 1: Mean

Mean of the output column

cgpa	package	Mean	Residual
6.7	4.5	7.3	-2.8
9.0	11.0	7.3	3.7
7.5	6.0	7.3	1.3
5.0	8.0	7.3	0.7



: Form a decision tree between input and residual 1

1) Similarity score is calculated

$$SS = \frac{(\text{sum of residual})^2}{\# \text{ of residual} + \lambda}$$

λ = regularization parameter

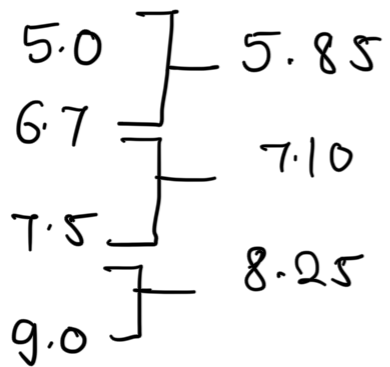
$$SS = \frac{(-2.8 + 3.7 - 1.3 + 0.7)^2}{4 + 0} = 0.02$$

2) To select root node

For this ss should be improved.
to do so,

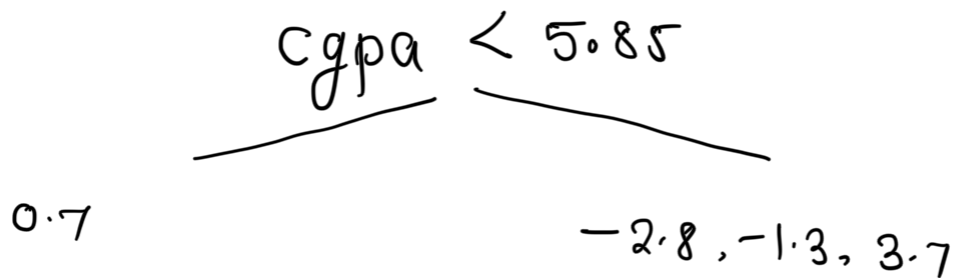
1) **data** is sorted, and mean is

— J^2 —
calculated for each split.



res. get splitted

Splitting criteria ① 5.85



2) Similarity is calculated for each node

$$SS_L = \frac{(0.7)^2}{1}$$

$$= 0.49$$

$$SS_R = \frac{(-2.8 - 1.3 + 3.7)^2}{3}$$

$$= 0.05$$

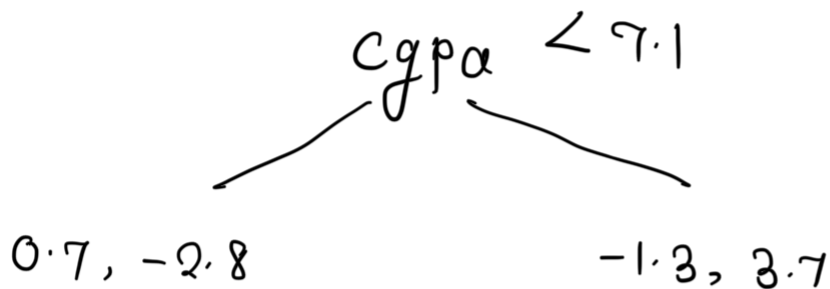
3) Similarity gain

$$\text{Gain} = (SS_L + SS_R) - SS_{\text{root}}$$

$$= 0.49 + 0.05 - 0.02$$

$$= 0.52$$

Splitting criteria ② $\rightarrow 7.1$



$$SS_L = \frac{(0.7 - 2.8)^2}{2 + 0}$$

$$= 2.20$$

$$SS_R = \frac{(-1.3 + 3.7)^2}{2 + 0}$$

$$= 2.88$$

$$\text{gain} = SS_L + SS_R - SS_{\text{root}}$$

$$\begin{aligned}
 J &= 2.20 + 2.88 - 0.02 \\
 &= 5.06
 \end{aligned}$$

Splitting criteria ③ $\rightarrow 8.25$

$$\begin{array}{ccc}
 & \text{cgpa} < 8.25 & \\
 \swarrow & & \searrow \\
 0.7, -2.8, -1.3 & & 3.7
 \end{array}$$

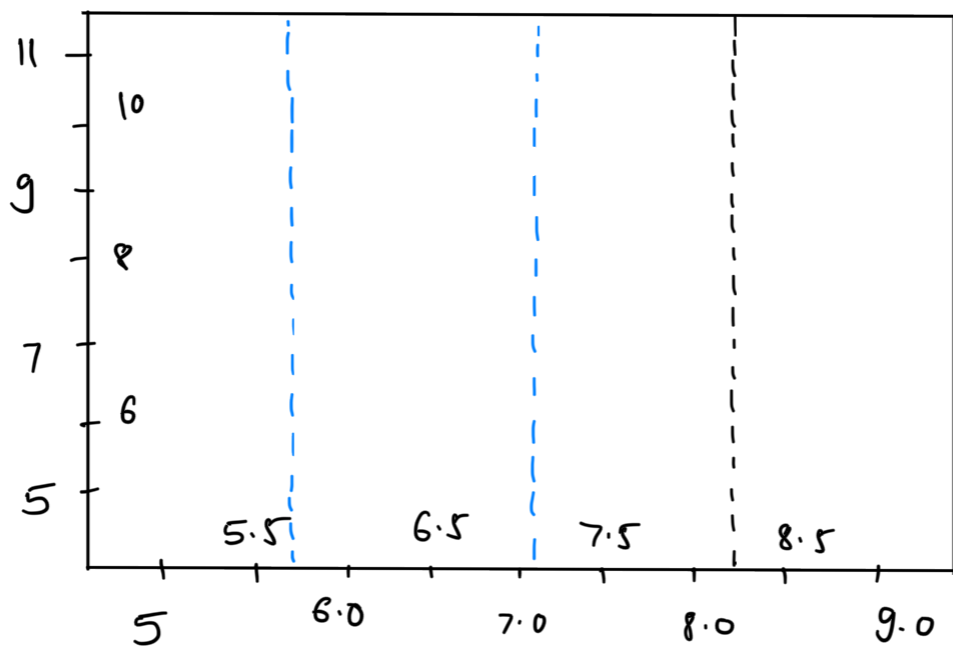
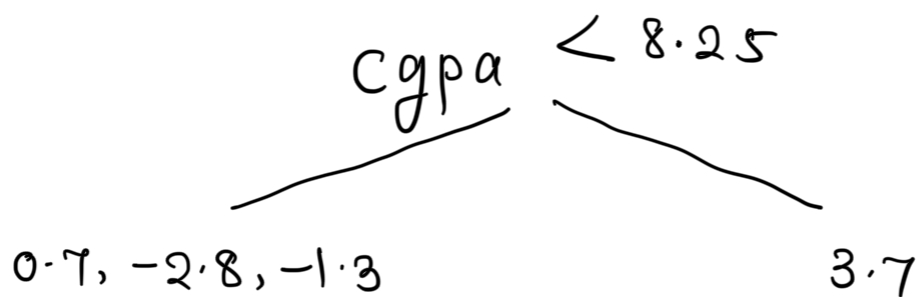
$$\begin{aligned}
 SS_L &= \frac{(0.7 - 2.8 - 1.3)^2}{3+0} \\
 &= 3.85
 \end{aligned}$$

$$\begin{aligned}
 SS_R &= \frac{(3.7)^2}{1} \\
 &= 13.69
 \end{aligned}$$

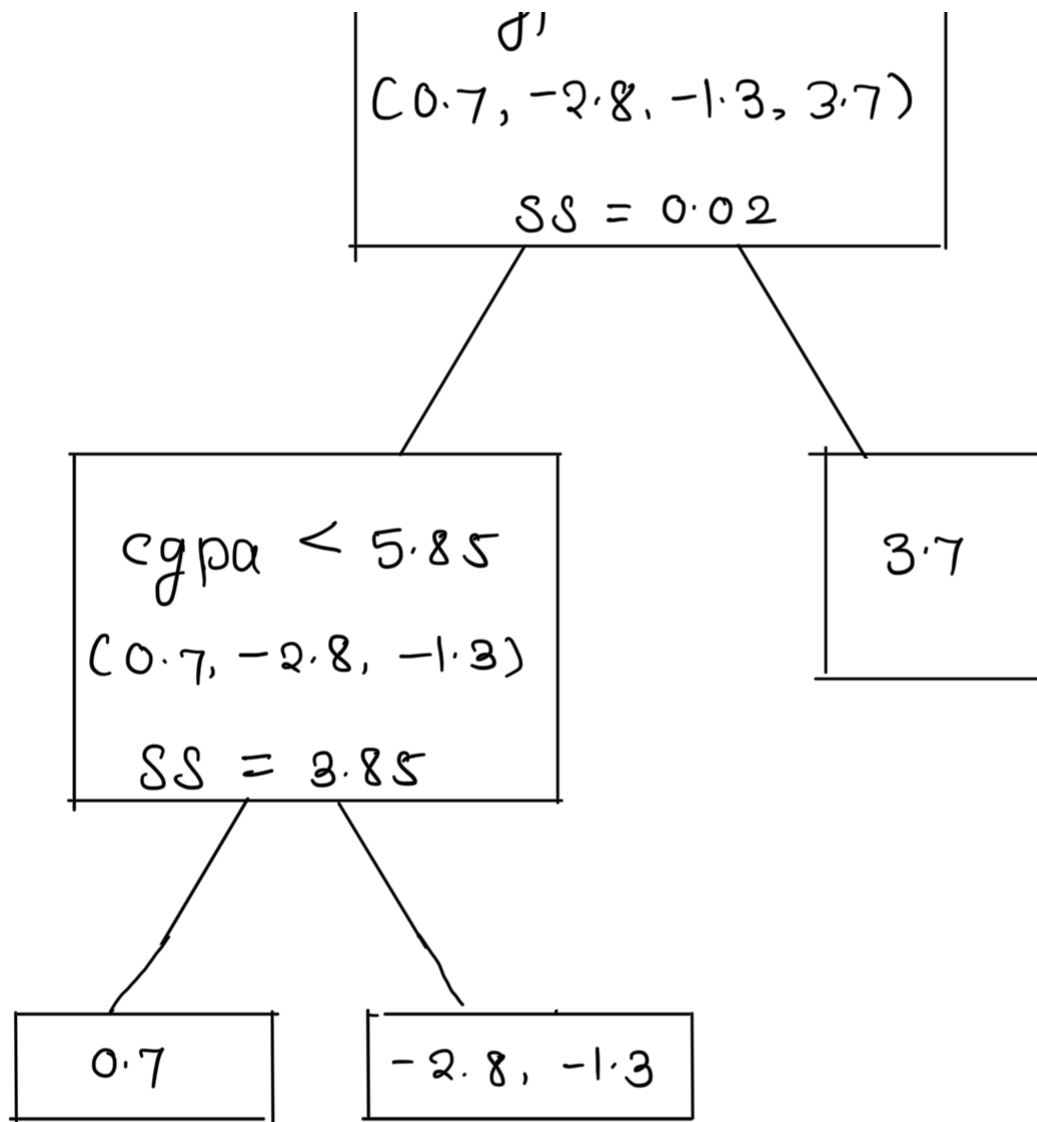
$$\begin{aligned}
 \text{gain} &= 3.85 + 13.69 - 0.02 \\
 &= 17.52
 \end{aligned}$$

We are getting max gain at splitting criteria 3

conclusion is in stage 2, the dt we are forming, has root cgra is
 $cgra < 8.25$



$cgra < 8.25$



$$SS_L = \frac{(0.7)^2}{1}$$

$$= 0.49$$

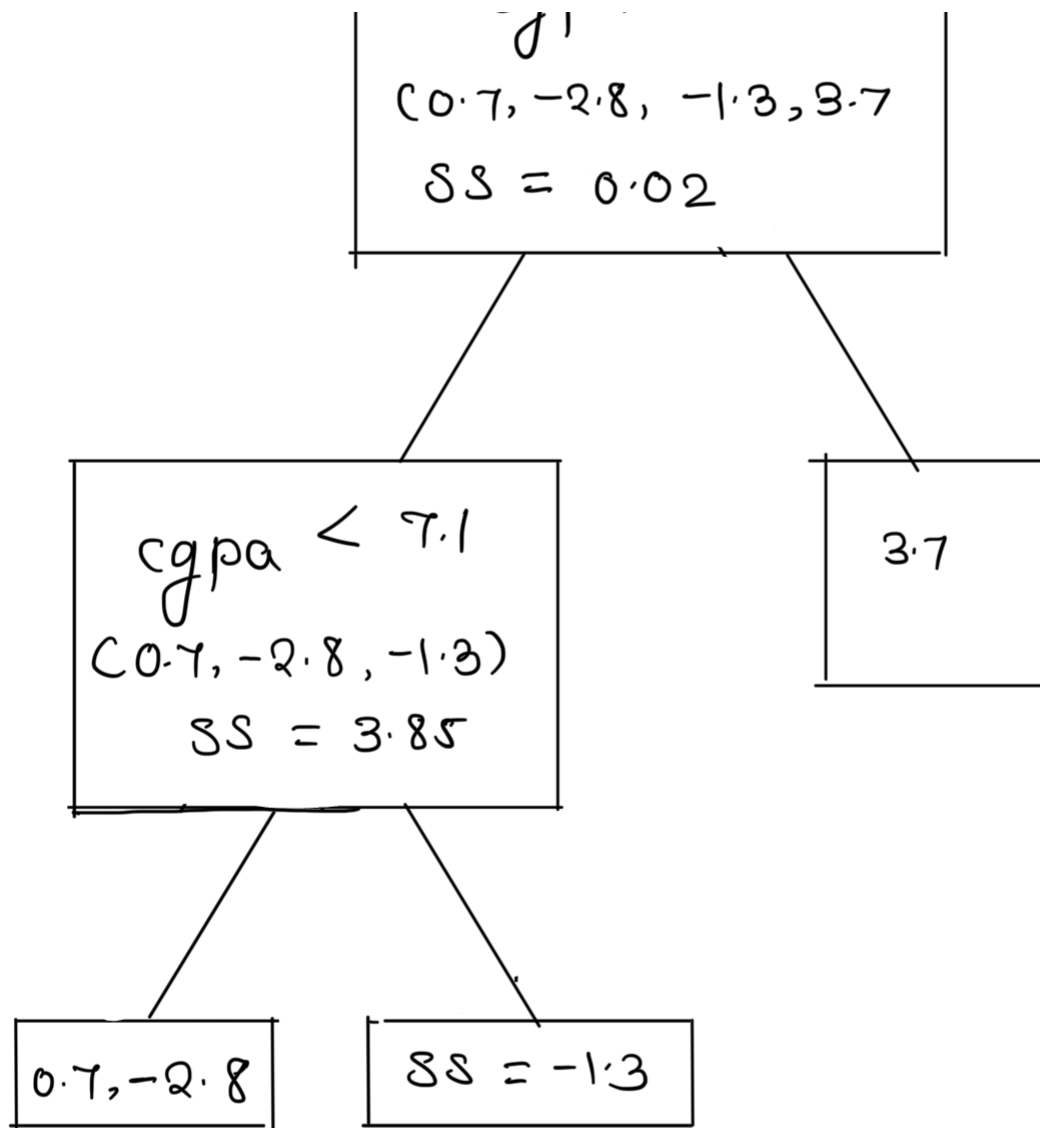
$$SS_R = \frac{(-2.8 - 1.3)^2}{2}$$

$$= 8.4$$

$$\text{Gain} = (SS_L + SS_R - SS_p) = 0.49 + 8.40 - 3.85$$

$$= 5.04$$

$$\text{cgpa} < 8.25$$



$$SS_L = \frac{(0.7 - 2.8)^2}{2}$$

$$= \frac{(-2.1)^2}{2}$$

$$= 2.20$$

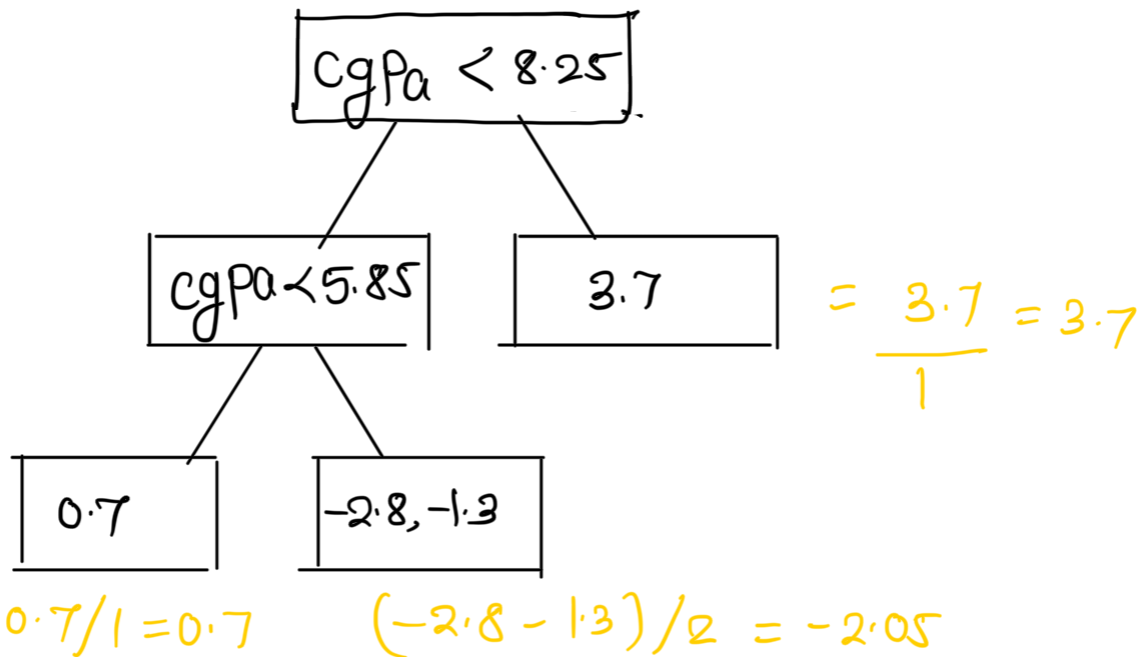
$$SS_R = \frac{(-1.3)^2}{1}$$

$$= 1.69$$

$$\text{Gain} = 2.20 + 1.64 - 3.85$$

$$= 0.04$$

Stage 3: Final decision tree

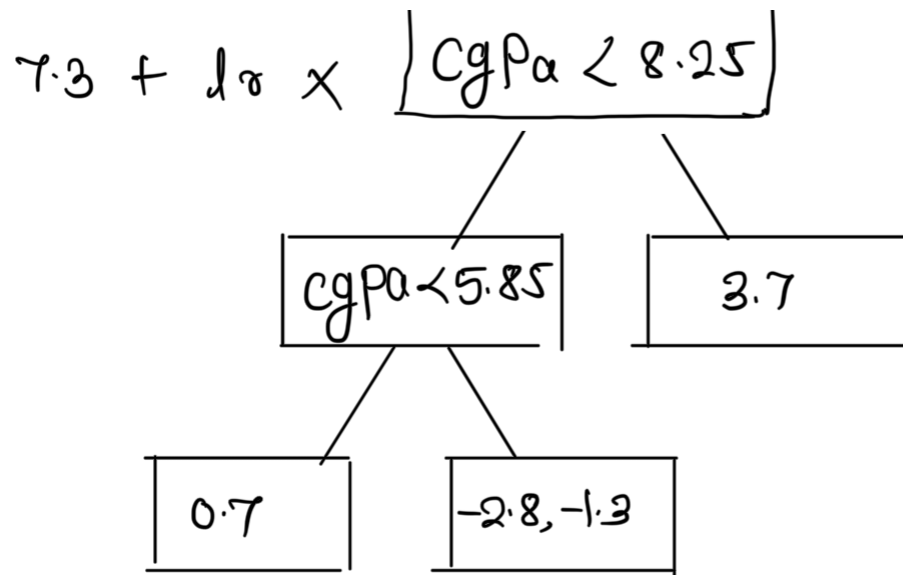


Output value for any leaf node

$$\text{output} = \frac{\text{Sum of residual}}{\# \text{ residual} + \lambda}$$

Now, prediction of Model 2

—————



$$1x = 0.3$$

cgpa	Pack	M1	res1	m2	res2
6.7	4.5	7.3	-2.8	6.69	-2.19
9.0	11.0	7.3	3.7	8.41	2.59
7.5	6.0	7.3	-1.3	6.69	-0.69
5.0	8.0	7.3	0.7	7.51	0.49

Model 2 Predictions

$$7.3 + 0.3 \times (-2.05)$$

$$7.3 + 0.3 \times 3.7$$

$$7.3 + 0.3 \times (-2.05)$$

$$7.3 + 0.3 \times 0.7$$

Stage 4 - Model 3

same Process as above.